### 2.3.6 Symplectic Integrators

A symplectic integrator is an approximation of the flow map that conserves the symplectic 2-form. Some first steps in the theory of symplectic integration, i.e. numerical methods explicitly designed to mimic the symplectic property of the flow map, were made by Vogalaere [98] (in 1956!) but this work went unnoticed. The first practical methods conserving the symplectic property were suggested by Ruth in 1983 [320] and followed by a number of works on a similar theme [71, 72, 129–131, 267, 353]. Later works, e.g. [132, 139, 214, 325, 398] were aimed at developing methods with higher order of accuracy or better understanding of the meaning of the symplectic property (we will discuss this aspect in the next chapter). Some symplectic integrators are found within existing families (like Runge-Kutta methods), but the most useful are typically obtained using a splitting and composition framework that allows us to build families of such methods.

In the sequel we will write $\boldsymbol{Z} = \mathscr{G}_h(\boldsymbol{z})$ to specify the starting point $\boldsymbol{z}$ and ending point $\boldsymbol{Z}$ of a step.

Alternatively, if we wish to emphasize the decomposition into positions and momenta, we write

$$\begin{bmatrix} \boldsymbol{Q} \\ \boldsymbol{P} \end{bmatrix} = \mathscr{G}_h\left(\begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{p} \end{bmatrix}\right).$$

Recall that, in terms of $\boldsymbol{q}$ and $\boldsymbol{p}$, the differential equations take the form

$$\dot{\boldsymbol{q}} = \nabla_p H, \qquad \dot{\boldsymbol{p}} = -\nabla_q H.$$

In molecular dynamics, the Hamiltonian is usually of the form

$$H = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q}),$$

with $\boldsymbol{M}$ a diagonal mass matrix, and we will concentrate on this case for the moment. In this case,

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \qquad \dot{\boldsymbol{p}} = -\nabla_q U(\boldsymbol{q}) \equiv \boldsymbol{F}(\boldsymbol{q}).$$

The following scheme is a slight modification of the Euler method.

$$\boldsymbol{Q} = \boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{P}, \tag{2.18}$$

$$\boldsymbol{P} = \boldsymbol{p} + h\boldsymbol{F}(\boldsymbol{q}). \tag{2.19}$$

The method is explicit: to advance the timestep, we use the second equation to compute $\boldsymbol{P}$ and then insert this in the first to get $\boldsymbol{q}$. Let the vectors $\boldsymbol{q}, \boldsymbol{p}, \boldsymbol{Q}, \boldsymbol{P}$ have $i$th